

Le Projet : Network & Security Sentinel

L'objectif est de transformer une machine Linux (Ubuntu Server ou Debian) en un tableau de bord textuel de surveillance.

1. Stack Technologique (CLI Only)

- **Log Management** : Fail2Ban pour la sécurité.
- **Monitoring** : Nmap (scans réseau) et MTR (diagnostic de route).
- **Automation** : Scripts Bash et Cron.
- **Visualisation CLI** : Btop ou Glances (mode serveur).

2. Architecture du Système

3. Étapes de Mise en Œuvre

Étape A : Durcissement et Sécurité (Fail2Ban)

Un ingénieur système doit protéger son accès. Configure Fail2Ban pour bannir automatiquement les IP qui tentent de forcer ton SSH.

```
root@ubuntu:~$ sudo apt update && sudo apt install fail2ban -y
sudo cp /etc/fail2ban/jail.conf /etc/fail2ban/jail.local
sudo systemctl start fail2ban
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  python3-pyasyncore python3-pyinotify whois
Suggested packages:
  mailx monit sqlite3 python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban python3-pyasyncore python3-pyinotify whois
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 496 kB of archives.
After this operation, 2572 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-pyasyncore all 1.0.2-2 [10.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 fail2ban all 1.0.2-3ubuntu0.1 [409 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-pyinotify all 0.9.6-2ubuntu1 [25.0 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 whois amd64 5.5.22 [51.7 kB]
Fetched 496 kB in 1s (794 kB/s)
Selecting previously unselected package python3-pyasyncore.
(Reading database ... 114047 files and directories currently installed.)
Preparing to unpack .../python3-pyasyncore_1.0.2-2_all.deb ...
Unpacking python3-pyasyncore (1.0.2-2) ...
Selecting previously unselected package fail2ban.
```

- **Défi :** Crée un script qui extrait les 10 dernières IP bannies et affiche leur pays d'origine via une API de géolocalisation en curl.

Étape B : Script de Monitoring Réseau (Bash)

Crée un script nommé `netcheck.sh` qui vérifie la disponibilité de tes serveurs critiques et logue les résultats.

```
root@ubuntu:~/netcheck$ cat netcheck
#!/bin/bash
TARGETS=("8.8.8.8" "google.com" "192.168.1.1")
LOGFILE="/var/log/network_sentinel.log"

echo "--- Check du $(date) ---" >> $LOGFILE
for ip in "${TARGETS[@]}; do
    if ping -c 1 $ip &> /dev/null; then
        echo "$ip est UP" >> $LOGFILE
    else
        echo "ALERT: $ip est DOWN" >> $LOGFILE
    fi
done
```

Étape C : Automatisation et Alertes

Planifie le scan avec crontab pour qu'il s'exécute toutes les 5 minutes.

```
GNU nano 7.2
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
*/5 * * * * /home/user/netcheck.sh >> /var/log/cron_netcheck.log 2>&1
```

tape D : Analyseur de Flux en Temps Réel

Utilise nload pour visualiser la bande passante et tcpdump pour capturer les paquets suspects sur un port spécifique :

```
root@ubuntu:~# ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 32:29:63:0c:0c:4e brd ff:ff:ff:ff:ff:ff
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1454 qdisc noqueue state DOWN mode DEFAULT group default
    link/ether 4a:b9:3f:c8:d3:a4 brd ff:ff:ff:ff:ff:ff
root@ubuntu:~# sudo tcpdump -i enp1s0 not port 22 -n
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp1s0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:37:47.273328 IP 172.30.1.2.40200 > 10.244.5.205.37062: Flags [P.], seq 3223147891:3223148024, ack 1220831251, win 498, options [nop,nop,TS val 3210864807 ecr 567562452], length 133
11:37:47.273623 IP 10.244.5.205.37062 > 172.30.1.2.40200: Flags [.], ack 133, win 502, options [nop,nop,TS val 567562474 ecr 3210864807], length 0
11:37:47.377903 IP 172.30.1.2.40200 > 10.244.5.205.37062: Flags [P.], seq 133:328, ack 1, win 498, options [nop,nop,TS val 3210864912 ecr 567562474], length 195
11:37:47.378041 IP 10.244.5.205.37062 > 172.30.1.2.40200: Flags [.], ack 328, win 502, options [nop,nop,TS val 567562578 ecr 3210864912], length 0
11:37:47.378186 IP 172.30.1.2.40200 > 10.244.5.205.37062: Flags [P.], seq 328:335, ack 1, win 498, options [nop,nop,TS val 3210864912 ecr 567562578], length 7
11:37:47.378519 IP 10.244.5.205.37062 > 172.30.1.2.40200: Flags [.], ack 335, win 502, options [nop,nop,TS val 567562579 ecr 3210864912], length 0
11:37:47.479028 IP 172.30.1.2.40200 > 10.244.5.205.37062: Flags [P.], seq 335:379, ack 1, win 498, options [nop,nop,TS val 3210865013 ecr 567562579], length 44
11:37:47.479245 IP 10.244.5.205.37062 > 172.30.1.2.40200: Flags [.], seq 379, win 502, options [nop,nop,TS val 567562680 ecr 3210865013], length 0
11:37:47.479546 IP 172.30.1.2.40200 > 10.244.5.205.37062: Flags [P.], seq 379:452, ack 1, win 498, options [nop,nop,TS val 3210865013 ecr 567562680], length 73
11:37:47.479655 IP 10.244.5.205.37062 > 172.30.1.2.40200: Flags [.], ack 452, win 502, options [nop,nop,TS val 567562680 ecr 3210865013], length 0
11:37:47.583319 IP 172.30.1.2.40200 > 10.244.5.205.37062: Flags [P.], seq 452:549, ack 1, win 498, options [nop,nop,TS val 3210865117 ecr 567562680], length 97
11:37:47.583474 IP 10.244.5.205.37062 > 172.30.1.2.40200: Flags [.], ack 549, win 502, options [nop,nop,TS val 567562784 ecr 3210865117], length 0
11:37:47.687101 IP 172.30.1.2.40200 > 10.244.5.205.37062: Flags [P.], seq 549:582, ack 1, win 498, options [nop,nop,TS val 3210865221 ecr 567562784], length 33
11:37:47.687357 IP 10.244.5.205.37062 > 172.30.1.2.40200: Flags [.], ack 582, win 502, options [nop,nop,TS val 567562888 ecr 3210865221], length 0
11:37:47.687379 IP 172.30.1.2.40200 > 10.244.5.205.37062: Flags [P.], seq 582:615, ack 1, win 498, options [nop,nop,TS val 3210865221 ecr 567562888], length 33
```

4. Ce que tu vas maîtriser

1. Gestion des logs : Manipulation de /var/log/auth.log et /var/log/syslog.

```
GNU nano 7.2 /var/log/auth.log
2025-02-10T22:04:55.272884+00:00 vm2 useradd[758]: new group: name=ubuntu, GID=1000
2025-02-10T22:04:55.272885+00:00 vm2 useradd[758]: new user: name=ubuntu, UID=1000, GID=1000, home=/home/ubuntu, shell=/bin/bash, from=none
2025-02-10T22:04:55.272887+00:00 vm2 useradd[758]: add 'ubuntu' to group 'adm'
2025-02-10T22:04:55.272888+00:00 vm2 useradd[758]: add 'ubuntu' to group 'cdrom'
2025-02-10T22:04:55.272889+00:00 vm2 useradd[758]: add 'ubuntu' to group 'sudo'
2025-02-10T22:04:55.272892+00:00 vm2 useradd[758]: add 'ubuntu' to group 'dip'
2025-02-10T22:04:55.272894+00:00 vm2 useradd[758]: add 'ubuntu' to group 'lxd'
2025-02-10T22:04:55.272895+00:00 vm2 useradd[758]: add 'ubuntu' to shadow group 'adm'
2025-02-10T22:04:55.272897+00:00 vm2 useradd[758]: add 'ubuntu' to shadow group 'cdrom'
2025-02-10T22:04:55.272898+00:00 vm2 useradd[758]: add 'ubuntu' to shadow group 'sudo'
2025-02-10T22:04:55.272899+00:00 vm2 useradd[758]: add 'ubuntu' to shadow group 'dip'
2025-02-10T22:04:55.272901+00:00 vm2 useradd[758]: add 'ubuntu' to shadow group 'lxd'
2025-02-10T22:04:55.272904+00:00 vm2 passwd[766]: password for 'ubuntu' changed by 'root'
2025-02-10T22:04:55.272905+00:00 vm2 chpasswd[776]: pam_unix(chpasswd:chauthtok): password changed for ubuntu
2025-02-10T22:04:55.273121+00:00 vm2 systemd-logind[802]: Watching system buttons on /dev/input/event0 (Power Button)
2025-02-10T22:04:55.273122+00:00 vm2 systemd-logind[802]: Watching system buttons on /dev/input/event1 (AT Translated Set 2 keyboard)
2025-02-10T22:04:55.273124+00:00 vm2 systemd-logind[802]: New seat seat0.
2025-02-10T22:04:55.289584+00:00 vm2 polkitd[792]: Loading rules from directory /etc/polkit-1/rules.d
2025-02-10T22:04:55.289586+00:00 vm2 polkitd[792]: Loading rules from directory /usr/share/polkit-1/rules.d
2025-02-10T22:04:55.289589+00:00 vm2 polkitd[792]: Finished loading, compiling and executing 6 rules
2025-02-10T22:04:55.289593+00:00 vm2 polkitd[792]: Acquired the name org.freedesktop.Policykit1 on the system bus
2025-02-10T22:04:57.532749+00:00 vm2 useradd[1132]: new user: name=kc-internal, UID=0, GID=0, home=/root, shell=/bin/bash, from=none
2025-02-10T22:05:01.717528+00:00 vm2 CRON[1163]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2025-02-10T22:05:01.719690+00:00 vm2 CRON[1163]: pam_unix(cron:session): session closed for user root
2025-02-10T22:05:24.650440+00:00 vm2 sshd[1169]: Server listening on :: port 22.
2025-02-10T22:05:32.570392+00:00 vm2 sshd[1171]: Accepted password for kc-internal from 10.244.0.131 port 34164 ssh2
2025-02-10T22:05:33.800434+00:00 vm2 sshd[1171]: Received disconnect from 10.244.0.131 port 34164:11: disconnected by user
2025-02-10T22:05:33.800603+00:00 vm2 sshd[1171]: Disconnected from user kc-internal 10.244.0.131 port 34164
2025-02-10T22:06:22.911964+00:00 vm2 sshd[1190]: Failed password for kc-internal from 10.244.0.131 port 52862 ssh2
2025-02-10T22:06:42.696877+00:00 vm2 sshd[1190]: Accepted password for kc-internal from 10.244.0.131 port 52862 ssh2
2025-02-10T22:06:43.912671+00:00 vm2 systemd-logind[802]: The system will halt now!
2025-02-10T22:06:43.925189+00:00 vm2 systemd-logind[802]: System is halting.
[ Read 202 lines ]
^G Help      ^O Write Out  ^W Where Is   ^X Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark  M-] To Brack
^X Exit      ^R Read File  ^N Replace    ^V Paste      ^J Justify    ^Y Go To Line M-E Redo     M-G Copy      ^O Where Was
```

2. **Scripts d'automatisation** : Utilisation de boucles, conditions et redirection de flux.
3. **Sécurité active** : Gestion des règles de pare-feu (iptables/ufw) via Fail2Ban.
4. **Analyse réseau** : Diagnostic profond sans aucun outil visuel type Wireshark.

5. Bonus pour l'ingénieur

Installe **Glances**. C'est un outil en Python qui permet de voir tout l'état du système (CPU, RAM, Process, IO Disk, Network) dans un seul terminal avec des codes couleurs.

```
root@ubuntu:~$ sudo apt install glances
glances

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  bit fonts-lyx isympy-common isympy3 libblas3 libdouble-conversion3 libgfortran5 libgraphite2-3 libharfbuzz0b libimagequant0 libjs-jquery-ui liblapack3 liblbfgsb0 liblcms2-2
  libopenjp2-7 libqhull-r8.0 libraqm0 libtk8.6 libwebpdemux2 libwebpmux3 libxft2 libxrender1 libxss1 lm-sensors python-matplotlib-data python3-appdirs python3-bottle
  python3-brotli python3-bs4 python3-contourpy python3-cssselect python3-cycler python3-decorator python3-defusedxml python3-fonttools python3-fs python3-html5lib
  python3-influxdb python3-kiwisolver python3-lxml python3-lz4 python3-matplotlib python3-mpmath python3-msgpack python3-numpy python3-olefile python3-pil python3-pil.imagetk
  python3-ply python3-psutil python3-pycryptodome python3-pysmi python3-pysnmp4 python3-pystache python3-scipy python3-soupsieve python3-sympy python3-tk python3-ufolib2
  python3-urllib3 python3-unicodedata2 python3-weencodings tk8.6-bit2.5 unicode-data x11-common
Suggested packages:
  bit-demo glances-doc python3-pymvml libjs-jquery-ui-docs liblcms2-utils tk8.6-fancontrol read-edid i2c-tools python-cycler-doc python3-genshi python-lxml-doc cm-super-minimal
  dvipng ffmpeg fonts-staypuft ghostscript gir1.2-gtk-3.0 inkscape python3-cairocffi python3-cairo python3-gobject python3-pyqt5 python3-sip
  python3-tornado texlive-extra-utils texlive-latex-extra python-mpmath-doc python3-gmpy2 gfortran python3-pytest python-pil-doc python-ply-doc python-scipy-doc
The following NEW packages will be installed:
  bit fonts-lyx glances isympy-common isympy3 libblas3 libdouble-conversion3 libgfortran5 libgraphite2-3 libharfbuzz0b libimagequant0 libjs-jquery-ui liblapack3 liblbfgsb0
  liblcms2-2 libopenjp2-7 libqhull-r8.0 libraqm0 libtk8.6 libwebpdemux2 libwebpmux3 libxft2 libxrender1 libxss1 lm-sensors python-matplotlib-data python3-appdirs python3-bottle
  python3-brotli python3-bs4 python3-contourpy python3-cssselect python3-cycler python3-decorator python3-defusedxml python3-fonttools python3-fs python3-html5lib
  python3-influxdb python3-kiwisolver python3-lxml python3-lz4 python3-matplotlib python3-mpmath python3-msgpack python3-numpy python3-olefile python3-pil python3-pil.imagetk
  python3-ply python3-psutil python3-pycryptodome python3-pysmi python3-pysnmp4 python3-pystache python3-scipy python3-soupsieve python3-sympy python3-tk python3-ufolib2
  python3-urllib3 python3-unicodedata2 python3-weencodings tk8.6-bit2.5 unicode-data x11-common
0 upgraded, 66 newly installed, 0 to remove and 0 not upgraded.
Need to get 56.2 MB of archives.
After this operation, 239 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
```

Résultat:

ubuntu (Ubuntu 24.04 64bit / Linux 6.8.0-101-generic)Uptime: 1:13:04

Intel Xeon E312xx (Sandy Bridge, IBRS update)

CPU ↓3.9%idle96.1%ctx_sw348MEM -29.4%active879Mswap -0.0%LOAD -1core

CPU [|||||]3.9%user1.0%irq0.0%inter167total1.86Ginacti502Mtotal1024M1 min0.93

MEM [|||||]29.4%system1.0%nice0.5%sw_int21used560Mbuffer134Mused05 min0.36

SWAP [|||||]0.0%iowait0.0%steal0.5%free1.31Gcached1.18Gfree1024M15 min0.14

NETWORK

enp1s01Kb2Kb

lo384b384b

TASKS 130 (202 thr), 2 run, 74 slp, 54 oth Threads sorted automatically by CPU consumption

DefaultGateway7ms

DISK I/O

vdas0W/s

vdas100

vdas400

vdas1500

vdas1600

FILE SYS

/ (vdas1)5.52G18.3G

Command ('e' to pin | 'k' to kill)

python3 /usr/bin/glances

node /opt/theia/node_modules/@theia/core/lib/node/messaging/ipc-bo

dockerd -H fd:// --containerd=/run/containerd/containerd.sock

node /opt/theia/browser-app/src-gen/backend/main.js /root --hostna

containerd

python3 /usr/bin/glances -s -B 127.0.0.1

python3 /usr/bin/fail2ban-server -xf start

multipathd -d -s

python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown

packagekitd

init

udisksd

systemd-resolved

systemd-journald

ModemManager

runtime-info-service

polkitd --no-debug

systemd-networkd

sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

systemd-logind

systemd-timesyncd

systemd-udev

rsyslogd -n -iNONE

@dbus-daemon --system --address=systemd: --nofork --nopidfile --sy

2026-03-06 11:42:13 UTC